Bisection vs Illinois:

A Comparison of Root-finding Methods

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Intro

These slides cover 3 root-finding methods:

1. Bisection
2. Regula falsi "method of false position"
3. Illinois method (Snyder, 1953; Dowell & Jarratt, 1971)

I will assume you know the concept of bisection, so that will be a quick review/reference point for us

Motivation

Why even do this?-

Well, why even do anything? If you're going to do something, do it well.

More importantly, when it's 3:30am and you're trying to speed through a calibration, each iteration takes years off your life and decades off your rapidly dwindling sanity

- You'd like to shrink the initial brackets, but when targeting strong asymmetries (e.g. size distributions), things don't behave as nicely as you'd like
- Widening the brackets adds 1, maybe 2 iters, but you are no longer a sane person that properly allocates your time

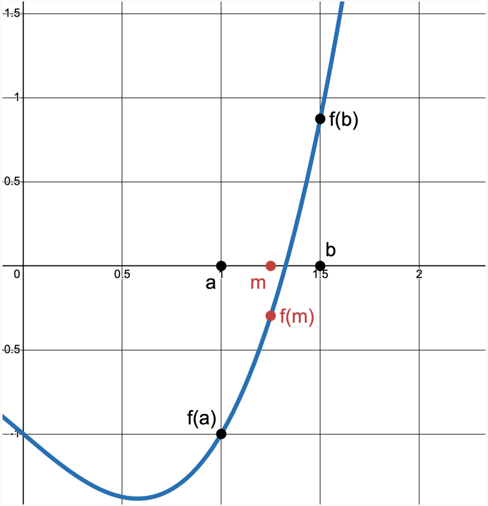
Bisection Review

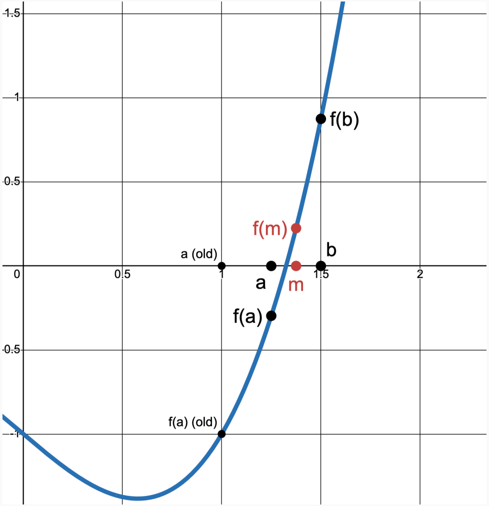
Suppose I wanted to find the root of $f(x)$, which is continuous on a sign-change bracket:

1. Pick initial brackets (call these a and b) where $f(a)$ and $f(b)$ have opposite signs
2. Find the midpoint of a and b, $m = \frac{a+b}{2}$ and take $f(m)$
3. If $f(m)$ has the same sign as $f(a)$, replace a with m
   - Or, replace b with m if $f(m)$ has the same sign as $f(b)$
4. Continue until $|f(m)| < desired\_tol$. Then m approximates the root

Next slide: example with $f(x) = x^3 - x - 1$ and initial $a = 1$ and $b = 1.5$

Bisection: Steps 1-3 (All graphs made with Desmos)





Bisection: Steps 4 - N

Now $m = 1.375$ becomes the new b and the process repeats until $f(m)$ is close to 0

- Determined by $|f(m)| < \epsilon$, where ϵ is sufficiently small and chosen by the modeler

This is a perfectly valid method, but we can make it faster with the “false position”

Regula Falsi

We use bisection, or various other numerical methods, because economics is hard

- (This is why Per Krusell's house is so big)
- What is simple is finding the root of a line

We can use the root of a line to better update our new a or b point:

1. As before, pick initial a and b where $f(a)$ and $f(b)$ have opposite signs
2. Instead of the midpoint, draw a line between $f(a)$ and $f(b)$
3. Call the root of this line c and take $f(c)$
    - I'm using c since this is no longer a midpoint, but call it m if you like
4. If $f(c)$ has the same sign as $f(a)$, replace a with c
    - Or, replace b with c if $f(c)$ has the same sign as $f(b)$
5. Continue until $|f(c)| < desired\_tol$. Then c is the root

Building the Line

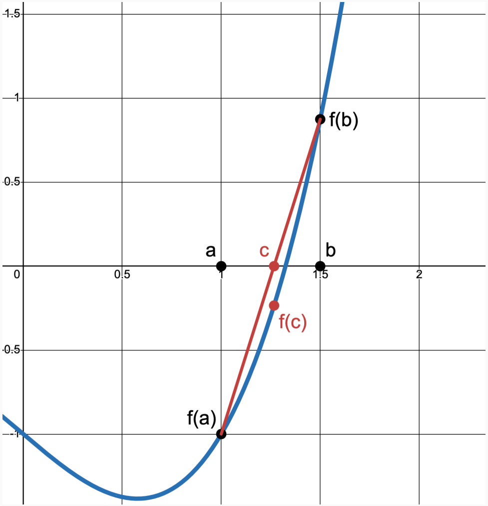
Linearly interpolate (and rearrange a bit) $f(a)$ and $f(b)$:

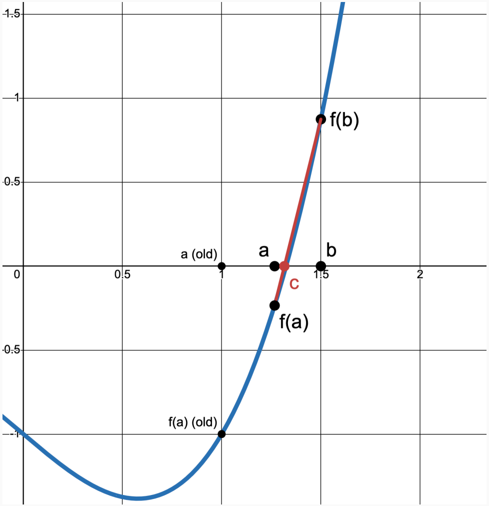
$$c = b - \frac{f(b) \times (b - a)}{f(b) - f(a)}$$

Optional: set a condition to use the smaller $|f(val)|$. If $|f(a)| < |f(b)|$, use:

$$c = {\color{red}a} - \frac{f({\color{red}a}) \times (b - a)}{f(b) - f(a)}$$

Regula Falsi: Steps 1-4

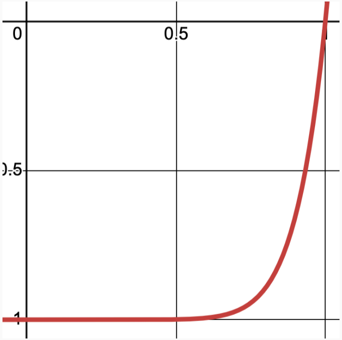




Issues?

Consider the case of $f(x) = x^{10} - 1$ on $[0, 1.3]$

- $f(x)$ is flat near 0 and blows up near 1.3
- Our line always crosses the axis around the left end
- $b = 1.3$ is retained forever and a crawls towards 1 (true root)



The Illinois Method

The Illinois method aims to correct this issue

- First as technical note by Snyder (1953), then formalized by Dowell & Jarratt (1971)

If sign $f(c)$ = sign $f(a)$, then the root is in $[c, b]$

- c is the new a; b remains the same

If this happens twice in a row, assign $f(b) = \frac{f(b)}{2}$ (same for $f(a)$ if opposite case)

- If it happens again on the next iter, halve again
- *Everything else remains the same as in regula falsi*
- Convergence guaranteed (halving does not change sign; stale bracket replaced)
  - ▷ Order of convergence ≈ 1.442 (Dowell & Jarratt, 1971)

Regula Falsi Failure and Illinois Correction for $f(x) = x^{10} - 1$

Regula Falsi:

```
iter=1  a=0.0  b=1.3  c=0.0942995953723274
iter=2  a=0.0942995953723274  b=1.3  c=0.18175887251907943
iter=3  a=0.18175887251907943  b=1.3  c=0.26287401252030435
iter=4  a=0.26287401252030435  b=1.3  c=0.3381051033222695
iter=5  a=0.3381051033222695  b=1.3  c=0.4078779165927524
iter=6  a=0.4078779165927524  b=1.3  c=0.47258315356239
iter=7  a=0.47258315356239  b=1.3  c=0.5325715106320142
iter=8  a=0.5325715106320142  b=1.3  c=0.5881445691706799
iter=9  a=0.5881445691706799  b=1.3  c=0.6395439707682159
iter=10  a=0.6395439707682159  b=1.3  c=0.6869431667412391
iter=11  a=0.6869431667412391  b=1.3  c=0.7304464366929381
iter=12  a=0.7304464366929381  b=1.3  c=0.7700987444968075
iter=13  a=0.7700987444968075  b=1.3  c=0.8059075090169241
iter=14  a=0.8059075090169241  b=1.3  c=0.8378738914244527
iter=15  a=0.8378738914244527  b=1.3  c=0.8660276320873561
iter=16  a=0.8660276320873561  b=1.3  c=0.8904572813509999
iter=17  a=0.8904572813509999  b=1.3  c=0.911328251308228
iter=18  a=0.911328251308228  b=1.3  c=0.9288846695992143
iter=19  a=0.9288846695992143  b=1.3  c=0.9434360517492835
iter=20  a=0.9434360517492835  b=1.3  c=0.9553339751921469
iter=21  a=0.9553339751921469  b=1.3  c=0.9649455723416893
iter=22  a=0.9649455723416893  b=1.3  c=0.9726296888054463
iter=23  a=0.9726296888054463  b=1.3  c=0.9787191336334624
```

Max iter hit at 100

Illinois:

```
iter=1  a=0.0  b=1.3  c=0.0942995953723274
iter=2  a=0.0942995953723274  b=1.3  c=0.18175887251907943
iter=3  a=0.18175887251907943  b=1.3  c=0.3330171567671213
iter=4  a=0.3330171567671213  b=1.3  c=0.5634423147022629
iter=5  a=0.5634423147022629  b=1.3  c=0.8463635731395354
iter=6  a=0.8463635731395354  b=1.3  c=1.074910177068493
iter=7  a=0.8463635731395354  b=1.074910177068493  c=0.9454923183277778
iter=8  a=0.9454923183277778  b=1.074910177068493  c=0.9828011093189349
iter=9  a=0.9828011093189349  b=1.074910177068493  c=1.0040954923602121
iter=10  a=0.9828011093189349  b=1.0040954923602121  c=0.9996755374546944
iter=11  a=0.9996755374546944  b=1.0040954923602121  c=0.9999940615677148
iter=12  a=0.9999940615677148  b=1.0040954923602121  c=1.000005704633256
iter=13  a=0.9999940615677148  b=1.000005704633256  c=0.9999999998475554
iter=14  a=0.9999999998475554  b=1.000005704633256  c=0.9999999999999961
```

My JMP D-StSt: Bisection vs Illinois

Bisection:

```
========================================================================
  METHOD 1: BISECTION    bracket [5, 10],  tol = 1e-08
========================================================================
 iter         pval         wage            C        f_dist
    1    5.0000000     0.428000     0.030432     -1.70e-01
    2   10.0000000     0.214000     0.568592     +4.69e-01
    3    7.5000000     0.285333     0.167265     +3.39e-02
    4    6.2500000     0.342400     0.077432     -8.26e-02
    5    6.8750000     0.311273     0.115721     -2.97e-02
    6    7.1875000     0.297739     0.139706     +5.76e-04
    7    7.0312500     0.304356     0.127275     -1.49e-02
    8    7.1093750     0.301011     0.133378     -7.28e-03
    9    7.1484375     0.299366     0.136514     -3.38e-03
   10    7.1679688     0.298550     0.138103     -1.41e-03
   11    7.1777344     0.298144     0.138903     -4.17e-04
   12    7.1826172     0.297942     0.139304     +7.92e-05
   13    7.1801758     0.298043     0.139104     -1.69e-04
   14    7.1813965     0.297992     0.139204     -4.48e-05
   15    7.1820068     0.297967     0.139254     +1.72e-05
   16    7.1817017     0.297980     0.139229     -1.38e-05
   17    7.1818542     0.297973     0.139242     +1.70e-06
   18    7.1817780     0.297976     0.139235     -6.06e-06
   19    7.1818161     0.297975     0.139238     -2.18e-06
   20    7.1818352     0.297974     0.139240     -2.43e-07
   21    7.1818447     0.297974     0.139241     +7.26e-07
   22    7.1818399     0.297974     0.139240     +2.42e-07
   23    7.1818376     0.297974     0.139240     -7.39e-10

  METHOD 1: BISECTION:  p = 7.1818376    23 solves total    146.48 seconds
```

Illinois:

```
========================================================================
  METHOD 2: ILLINOIS    bracket [5, 10],  tol = 1e-08
========================================================================
 iter         pval         wage            C        f_dist
    1    5.0000000     0.428000     0.030432     -1.70e-01
    2   10.0000000     0.214000     0.568592     +4.69e-01
    3    6.3285719     0.338149     0.081606     -7.64e-02
    4    6.8432961     0.312715     0.113480     -3.26e-02
    5    7.2293715     0.296015     0.143187     +4.86e-03
    6    7.1793218     0.298078     0.139033     -2.56e-04
    7    7.1818209     0.297975     0.139239     -1.70e-06
    8    7.1818540     0.297973     0.139241     +1.67e-06
    9    7.1818376     0.297974     0.139240     -6.44e-12

  METHOD 2: ILLINOIS:  p = 7.1818376    9 solves total    57.87 seconds
```

(Using xtol ($|b - a| < \epsilon$) makes the difference even stronger)

Other Methods?

Many other weights (or functions of weights) exist, along with a multitude of completely different methods, which may be faster

The Illinois method has three benefits in one: speed, [relative] ease of learning, and guaranteed to work (again, on a continuous on a sign-change bracket)

References

Dowell, Mark & Peter Jarratt (1971). **"A Modified Regula Falsi Method for Computing the Root of an Equation".** In: *BIT* 11.2, pp. 168–174. DOI: `10.1007/BF01934364`.

Snyder, James N. (1953). ***Inverse interpolation, a real root of*** $f(x) = 0$. ILLIAC I Library Routine H1-71. University of Illinois Digital Computer Laboratory, p. 4.

Graphs: https://www.desmos.com/calculator